

Erdős Problem #983

EP-983

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $n \geq 2$ and $\pi(n) < k \leq n$. Let $f(k, n)$ be the smallest integer r such that in any $A \subseteq \{1, \dots, n\}$ of size $|A| = k$ there exist primes $p_1, \dots, p_r$ such that $> r$ many $a \in A$ are only divisible by primes from $\{p_1, \dots, p_r\}$.

Is it true that $2\pi(n^{1/2}) - f(\pi(n) + 1, n) \rightarrow \infty$ as $n \rightarrow \infty$?

In general, estimate $f(k, n)$, particularly when $\pi(n) + 1 < k = o(n)$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>